BUS 123 · MATH-M06-L01 · Inventory, Overhead & Cost of Goods Sold

Course: Solving Business Problems with Technology · Fall 2026 **Track:** MATH · **Module:** M06 · **Lesson:** L01 **Case Study Company:** Tidal Goods Co.

1 · CONNECT TO PRIOR KNOWLEDGE

Earlier in the math track, you learned how retailers use markup to set selling prices from a known cost — how a 50% markup on a $40 item produces a $60 retail price. But markup math assumes you already know the cost of that item with precision.

This module goes one level deeper: **what exactly is the cost of the item you just sold?** When a business carries hundreds of units bought at different prices across different weeks, the answer is not obvious. The method you choose — FIFO, LIFO, or Weighted Average Cost — changes your reported profit, your tax bill, and the asset value on your balance sheet.

Layered on top of product cost are **overhead expenses**: rent, utilities, insurance, and indirect labor that keep the business running but cannot be traced to any single unit sold. Together, these elements feed the **Cost of Goods Sold (COGS)** calculation — the single most important cost line on a product business's income statement.

2 · CORE CONCEPTS

Part 1 — Inventory Valuation

Why Valuation Matters

Tidal Goods Co. is a coastal outdoor retailer selling paddleboards, surf gear, and accessories. Over a single quarter, Tidal places three purchase orders for the same paddleboard model as supplier prices shift due to tariff pressures and shipping costs:

Purchase Order	Unit Cost	Quantity
PO-101 (Feb)	$210/unit	10 units
PO-102 (Mar)	$230/unit	10 units
PO-103 (Apr)	$250/unit	10 units

Tidal sells 20 paddleboards during the quarter. The question every accountant must answer: which 20 units' costs flow to the income statement, and which 10 remain on the balance sheet? This is a **formal accounting policy decision** with real financial consequences.

The Four Valuation Methods

Specific Identification traces each physical unit to its exact purchase cost. This works for unique, high-value items — custom boats, fine jewelry, artwork — but is impractical for any business selling fungible goods in volume.

First-In, First-Out (FIFO) assumes the oldest inventory is sold first. Tidal's oldest boards (at $210) flow to COGS before the newer, more expensive ones. During inflation, FIFO produces **lower COGS** and **higher gross profit**. Ending inventory reflects the most recent, higher prices.

Last-In, First-Out (LIFO) assumes the newest inventory is sold first. The most recently purchased boards ($250) flow to COGS first. During inflation, LIFO produces **higher COGS** and **lower taxable income** — a cash flow advantage. However, LIFO is **prohibited under IFRS**, making it unavailable to companies that report under international rules.

Weighted Average Cost (WAC) calculates a single blended unit cost by dividing total cost available by total units available. It smooths out price fluctuations and is popular for commodity-type goods.

WAC Unit Cost Formula
`WAC Unit Cost = Total Cost of Goods Available / Total Units Available`
Tidal: ($2,100 + $2,300 + $2,500) / 30 units = $230.00/unit
COGS (WAC): 20 units × $230.00 = $4,600

The Numbers Side by Side

Method	COGS	Ending Inventory
FIFO	10×$210 + 10×$230 = **$4,400**	$2,500 (10 units × $250)
LIFO	10×$250 + 10×$230 = **$4,800**	$2,100 (10 units × $210)
WAC	20 × $230.00 = **$4,600**	$2,300 (10 units × $230)

The $400 spread between FIFO and LIFO COGS is not rounding — it is a direct result of the accounting policy chosen. Both methods describe the same 20 boards sold from the same physical stockroom. This is why the choice of inventory method is a **strategic management decision**, not just a bookkeeping detail.

LIFO and International Reporting
LIFO may reduce taxes in the short term, but any US company that wants to report under IFRS (required for most international stock exchanges) cannot use LIFO. If Tidal ever pursued a global expansion or international listing, it would need to restate all historical financials — a costly and complex process.

Part 2 — Cost of Goods Sold (COGS)

The Core Formula

COGS is the total direct cost of the products a business sold during a period. For a retailer like Tidal, this means the purchase cost of every item that left the store or shipped to a customer.

Beginning Inventory + Purchases – Ending Inventory = COGS

In plain English: start with what you had, add what you bought, subtract what is left, and the result is what you sold.

Tidal Goods Co. — Q1 2026

Line Item	Amount
Beginning Inventory	$12,400
+ Purchases	$38,600
= Goods Available	$51,000
– Ending Inventory	($9,800)
= COGS	**$41,200**
Revenue	$110,000
= Gross Profit	$68,800
Gross Margin %	62.5%

Why Ending Inventory Accuracy Is Critical

Notice that ending inventory **directly reduces COGS**. If the physical count at period end is wrong — even by a small amount — COGS is wrong, which means gross profit is wrong, which means net income is wrong. Worse, that error does not disappear: the ending inventory of this period becomes the beginning inventory of the next period, carrying the error forward.

Common Mistake — Overstated Ending Inventory
If Tidal's count overstates ending inventory by $5,000:
- Period 1: COGS understated by $5,000 → net income OVERSTATED by $5,000
- Period 2: Beginning inventory now too high → COGS overstated → net income UNDERSTATED
The errors cancel over two periods — but both periods reported false numbers.

Part 3 — Margin vs. Markup

Both margin and markup measure profitability from the same transaction. They use the same inputs — revenue and cost — but **divide by different denominators**. Confusing them is one of the most common and costly errors in retail pricing.

Measure	Formula	Tidal Example ($80 price, $48 cost)	Result	Used By
Gross Margin	`(Revenue − Cost) / Revenue`	($80 – $48) / $80	**40.0%**	Finance, investors
Markup	`(Revenue − Cost) / Cost`	($80 – $48) / $48	**66.7%**	Buyers, sales teams

A buyer who says "we mark up everything 100%" means the selling price is double the cost. That sounds like 100% profit — but the gross margin is only **50%**, because margin divides by the larger selling price. A pricing team that confuses the two will systematically underprice every product in the catalog.

Quick Check — Converting Between Them

- Given markup of 66.7%: `Margin = Markup / (1 + Markup)` = 0.667 / 1.667 = **40%**
- Given margin of 40%: `Markup = Margin / (1 − Margin)` = 0.40 / 0.60 = **66.7%**

Part 4 — Overhead Allocation

What Is Overhead?

Overhead costs are indirect, recurring expenses required to keep the business operational that **cannot be traced to any single unit sold**.

Tidal's monthly overhead:

Category	Monthly Amount
Rent	$9,000
Utilities	$4,200
Insurance	$3,800
Indirect labor	$3,500

Category	Monthly Amount
Maintenance	$1,500
Total	**$22,000**

Whether Tidal sells one board or one thousand, these costs exist.

Allocating Across Departments

Tidal operates two departments: a retail Surf Shop (2,400 sq ft) and an Online Warehouse (1,600 sq ft). Most overhead categories are allocated by **square footage**, because those costs increase with physical space.

Department	Sq Ft	Allocation Basis	Overhead Allocated
Surf Shop	2,400 / 4,000 = 60%	Sq footage	60% × $22,000 = **$13,200**
Warehouse	1,600 / 4,000 = 40%	Sq footage	40% × $22,000 = **$8,800**
Total	100%		**$22,000**

The Overhead Rate

`Overhead Rate = Total Overhead / Revenue = $22,000 / $110,000 = 20.0%`

A 20% overhead rate means that for every dollar of revenue Tidal earns, **20 cents goes to keeping the lights on** before COGS is even considered. When this rate rises unexpectedly, management investigates: Did rent increase? Did revenue drop while fixed costs stayed flat?

Choosing the Right Allocation Basis
- Square footage — works well for space-based costs (rent, utilities, insurance)
- Revenue share — works better for costs driven by sales volume (commissions, processing fees)
- Direct labor hours — works best for labor-intensive production overhead

The key: the basis should reflect what actually drives the cost.

3 · FORMULA REFERENCE

Formula	Expression	Tidal Goods Example
COGS	`Beg Inv + Purchases – End Inv`	$12,400 + $38,600 – $9,800 = **$41,200**

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Formula	Expression	Tidal Goods Example
Gross Profit	`Revenue – COGS`	$110,000 – $41,200 = **$68,800**
Gross Margin %	`Gross Profit / Revenue`	$68,800 / $110,000 = **62.5%**
Markup %	`(Revenue – Cost) / Cost`	($80 – $48) / $48 = **66.7%**
Margin %	`(Revenue – Cost) / Revenue`	($80 – $48) / $80 = **40.0%**
WAC Unit Cost	`Total Cost Avail / Total Units`	$6,900 / 30 = **$230.00**
FIFO COGS	`Oldest units × oldest cost`	10×$210 + 10×$230 = **$4,400**
Dept OH %	`Dept Sq Ft / Total Sq Ft`	2,400 / 4,000 = **60%**
OH Allocated	`Total OH × Dept %`	$22,000 × 60% = **$13,200**
OH Rate	`Total Overhead / Revenue`	$22,000 / $110,000 = **20.0%**

4 · CHECK YOUR UNDERSTANDING

Answer all seven questions before class. Answers appear at the end of this document.

1. Tidal Goods Co. had beginning inventory of $14,000, made purchases of $32,000 during the quarter, and ended with inventory of $11,500. Calculate COGS.
2. Using the paddleboard purchase orders in Part 1 (10 units at $210, 10 at $230, 10 at $250), calculate the COGS if Tidal sells 15 units using the **FIFO** method.
3. Using the same purchase orders, calculate the Weighted Average Cost per unit and the COGS for selling 15 units under **WAC**.
4. Tidal sells a wetsuit for $120. The landed cost is $72. Calculate both the **gross margin percentage** and the **markup percentage**.
5. Tidal's overhead for the month is $18,000. Revenue is $90,000. Calculate the overhead rate. If revenue drops to $75,000 next month but overhead stays flat, what is the new overhead rate?
6. Tidal's Surf Shop occupies 3,000 sq ft and the Warehouse occupies 2,000 sq ft. Monthly rent is $10,000. How much rent should be allocated to each department?

7 Tidal's ending inventory count is accidentally overstated by $3,000 this quarter. Explain what happens to COGS and net income this period, and what happens in the following period as a result.

Answer Key · Check Your Understanding

#	Answer
1	COGS = $14,000 + $32,000 – $11,500 = **$34,500**
2	FIFO — sell oldest first: 10 units × $210 + 5 units × $230 = $2,100 + $1,150 = **$3,250**
3	WAC unit cost = ($2,100 + $2,300 + $2,500) / 30 = **$230.00**. COGS = 15 × $230.00 = **$3,450**
4	Gross Margin = ($120 – $72) / $120 = **40.0%**. Markup = ($120 – $72) / $72 = **66.7%**
5	Current: $18,000 / $90,000 = **20.0%**. If revenue drops: $18,000 / $75,000 = **24.0%** — same overhead, less revenue to absorb it.
6	Surf Shop: 3,000 / 5,000 = 60% × $10,000 = **$6,000**. Warehouse: 2,000 / 5,000 = 40% × $10,000 = **$4,000**.
7	This period: ending inventory overstated by $3,000 → COGS understated → net income **overstated** by $3,000. Next period: beginning inventory too high → COGS overstated → net income **understated** by $3,000. The errors offset over two periods, but both periods reported incorrect figures.

5 · KEY VOCABULARY

Term	Definition
Cost of Goods Sold (COGS)	The direct cost of products sold during a period. Calculated as: Beginning Inventory + Purchases – Ending Inventory.

Term	Definition
FIFO	First-In, First-Out. Inventory method that assumes the oldest inventory is sold first. Produces lower COGS and higher profit during inflationary periods.
LIFO	Last-In, First-Out. Inventory method that assumes the newest inventory is sold first. Produces higher COGS and lower taxable income. Prohibited under IFRS.
Weighted Average Cost (WAC)	Inventory method that calculates a single blended unit cost by dividing total cost available by total units available.
Gross Margin	Profit expressed as a percentage of revenue: `(Revenue – Cost) / Revenue`. Used by finance teams and investors.
Markup	Profit expressed as a percentage of cost: `(Revenue – Cost) / Cost`. Used by purchasing and sales teams to set prices from known costs.
Overhead	Indirect, recurring costs that keep the business operational but cannot be traced to any single unit sold.
Overhead Allocation	The process of distributing shared overhead costs across departments using a practical basis such as square footage or revenue share.
Overhead Rate	Total overhead costs as a percentage of revenue. A rising rate signals cost control issues or falling revenue.
Goods Available for Sale	Beginning Inventory + Purchases. The total inventory that could have been sold; COGS cannot exceed this figure.

Bring to Class

Be ready to turn Tidal Goods Co.'s raw transaction information into a management report. The class activity will ask you to build a COGS schedule, compare inventory methods, calculate margin versus markup, and allocate overhead in a way a manager could explain.